Autonomous Dry Floor Cleaning Robot

Introduction

The proposed project involves the design and realization of a low-cost autonomous dry floor cleaning robot for small indoor spaces. In Sri Lankan homes, boarding facilities, offices, and classrooms, cleaning the floors is a repetitive and time-consuming activity. Although robotic cleaning equipment is available in the global market, its application in local contexts is limited owing to factors such as high cost, poor serviceability, and lack of optimization for common local floor conditions such as tile and cement floors.

Currently, cleaning activities are either carried out using traditional methods or using traditional vacuum cleaning equipment, both of which require continuous involvement of humans. On the other hand, available robotic cleaning equipment in local contexts are marketed as high-end imported equipment and are therefore beyond the purchasing capability of common users. Furthermore, low-cost robotic cleaning equipment available in the market offer limited functionality and poor cleaning performance.

To address the limitations in available equipment, this project proposes the realization of a cost-optimized robotic cleaning equipment that is optimized for local conditions..

Approach

The proposed system will be developed as an integrated embedded product consisting of mobility, sensing, cleaning, and power subsystems.

The robot will operate using a differential drive mechanism with two independently controlled wheel motors and a caster wheel for stability. Autonomous navigation will be achieved using a combination of obstacle detection sensors, bumper switches, and cliff sensors to prevent collisions and falls. A simplified reactive navigation algorithm will be implemented to enable the robot to move efficiently within confined indoor spaces without requiring complex mapping systems.

The cleaning mechanism will consist of a rotating side brush to collect debris from edges and direct it toward a central intake. A suction-assisted cleaning system will be implemented using a compact blower, allowing the robot to collect dust, hair, and small particles into a removable dustbin. A filter mechanism will be included to prevent dust from reaching the blower and ensure reliable operation.

The electronic system will be designed using a custom PCB integrating the microcontroller, motor drivers, sensor interfaces, power regulation, and protection circuitry. Unlike basic prototype systems, the design will avoid reliance on development boards and instead utilize standalone integrated circuits.

The robot will be powered by a rechargeable battery system with appropriate protection and charging circuitry. The system will also include user interface elements such as status indicators, control buttons, and basic fault indication mechanisms.

Proposed Features of the System

- Autonomous navigation using obstacle detection and reactive movement logic
- Cliff detection to prevent falling from elevated surfaces
- Side brush and suction-assisted cleaning mechanism
- Removable dustbin with filtering system
- Rechargeable battery-powered operation
- Custom PCB integrating control, sensing, and power subsystems
- Motor control using dedicated driver circuits
- Fault detection and basic user feedback via LEDs and a buzzer
- Compact and modular mechanical design for easy maintenance

Development Plan

Initially, the system will be developed as a prototype targeted at small indoor spaces such as rooms and offices. The development will follow a staged approach.

First, a basic mobile platform will be developed to validate motor control and movement. Next, sensing capabilities, including obstacle detection and cliff detection, will be implemented and tested. The cleaning subsystem, including the brush and suction mechanism, will then be developed and evaluated independently.

Finally, the complete system will be assembled and tested under real operating conditions to evaluate cleaning performance, battery life, navigation behavior, and reliability.

Conclusion

In conclusion, the proposed autonomous dry floor-cleaning robot presents a practical and product-oriented solution to a common everyday problem. While similar products exist in the market, this project differentiates itself by focusing on affordability, local suitability, and simplified yet effective design.